TECHNICAL SPECIFICATION FOR SILICON RUBBER COMPOSITE INSULATORS

1 SCOPE:

This specification covers design, manufacture, testing at manufacturer's works before dispatch and supply of Silicone Rubber Composite Insulators as per the requirements given in Section-IV of the Specification.

2. STANDARDS

The manufacturer should confirm the product with following Indian Standard, International Standards containing latest revisions, amendments, changes adopted.

Sl. No.	Indian Standards	Title	International Standards
1	IS:209-1992	Specifications for Zinc	BS:3436
2 3	IS:406-1991	Method of Chemical Analysis of Slab Zinc Composite insulators for A.C. Over head Power lines with a nominal voltage greater than 1000V	BS:3436 IEC: 61109-1992
4	IS Part Part (II) - 1991 Part (II) - 1991	Methods of High Voltage Testing.	IEC 60060-1
5	IS : 2486 Part I – 1993 Part II – 1989 Part – III 1991	Specification for Insulator fittings for Over Head Power Lines with a nominal voltage greater than 1000V General requirements and Tests. Dimensional Requirements. Locking Devices.	IEC : 575 BS-3288 IEC-6020 IEC-60372
6	IS : 2629 – 1990	Recommended Practice for Hot dip galvanization for iron and steel.	ISO-1461 (E)
7	IS : 2633-1992	Testing of Uniformity of Coating of zinc coated articles.	
8	IS –6745-1990	Determination of weight of Zinc Coating on Zinc coated iron and steel articles.	BS : 443-1969 ISO 1460-1973
9	IS : 8263-1990	Methods of RI test of HV insulators	IEC-60437 NEMA Publication No. 07/1964 CISPR
10	IS : 8269-1990	Methods for Switching Impulse test on HV insulators.	IEC-60506
11		Salt Fog Pollution Voltage Withstand Test.	IEC – 60507
12		Guide for the selection of insulators in respect of polluted conditions.	IEC – 60815

13		Tests or insulators of Ceramic material or glass or glass for overhead lines with a nominal voltage greater than 1000V	IEC - 60363

Analysis of Material Properties:

(i) Housing material

(1)	Resistance to tracking & erosion	IEC 60587
(2)	Tear strength KN/m	ASTM D 624-B
(3)	Volume resistivity	IEC 60093
(4)	Resistance to Weathering & UV	ASTM G 53-96
(5)	Arc Resistance	ASTM-495-1973
(6)	Specific gravity	ASTM D 792
(7)	Dielectric strength in KV/mm	ASTM D 149/IEC 60243
(8)	Hardness -Shore(A)	ASTM D 2204
(9)	Ultimate elongation	ASTM D 412
(10)	Tensile Strength	ASTM D 412
(11)	Living Oxygen Index	ASTM D 2863

(ii) On Core Material

(1)	Dye penetration test:	IEC 61109
(2)	Percentage of Mass:	ASTM D 2584
(3)	Hardness Test (Bacol):	ASTM D 2584
(4)	Water absorption:	IEC 61109, ASTM D570
(5)	Specific gravity:	ASTM D792
(6)	Brittle fracture resistance test:	as per relevant IEC/ASTM.

However in an event of supply of insulators conforming to standards other than specified, the bidder shall confirm in his bid that these standards are equivalent to those specified. In case of award, salient features of comparison between the standards proposed by the bidder and those specified in this document will be provided by the supplier to establish equivalence.

3. ATMOSPHERIC CONDITIONS:

i)	(a)	Maximum temperature of air in shade	55 Deg C
	(b)	Maximum average temperature	40 Deg C
ii)		Minimum temperature of air in shade	7.5 Deg C
iii)		Maximum temperature of air in shade attainable by an object exposed to sun.	70 Deg C
iv)		Maximum humidity	90%
v)		Average number of thunder-storm days per annum	50
vi)		Average number of rainy days per annum	90
vii)		Average rain fall per annum	1500mm
viii)		Number of months of tropical monsoon conditions	4 months.
ix)		Maximum wind pressure	205 Kg/Sq. Metres.
x)		Altitude not exceeding above MSL.	1000 Metres
xi)		Level of pollution	very heavily polluted

4. SYSTEM CONDITIONS.

Electrical system Data:

a)	System Voltage (KV rms)	:	132	220	400
b)	Max. Voltage (KV rms)	:	145	245	420
c)	Lightning Impulse withstand Voltage (Dry & Wet) (KVP)	:	650	1050	1550 (1425 for GIS)
d)	Power frequency withstand Voltage (wet) (KVP)	:	275	460	650
e)	Short circuit level and duration:		40 KA per one second.	50 KA per one second.	63 KA per one second

The materials offered shall be suitable for operation in tropical climate in which they will be subject to the full rays of the sun and inclement weather and shall be able to withstand wide range of temperature variations. The humidity may be as high as 100% during rainy season and as low as 10% during dry season. The prevailing weather conditions are given above.

Reference atmospheric conditions for which insulator characteristics shall be expressed for the purpose of comparison are given below.

Ambient temperature	20 Deg C
Barometric pressure	1013 Millibars
Absolute humidity	11 gms. Of water per cubic Metre corresponding to 63 percent relative humidity at 20 Deg.C

Tests for the purpose of this specification may be carried out under conditions naturally obtaining at the time of the test. The barometric pressure, air temperature and humidity shall be recorded for the purpose of corrections.

5.0 GENERAL REQUIRMENT

The design, manufacturing, processes, tolerances and inspection of composite insulators shall confirm to the following.

5.1 Language and units.

5.1.1 All correspondence, literature, drawings and markings shall be in the English language.

5.1.2 Dimensioning shall be in the SI (Metric System) units. Manufacture should mention the standard adopted for Dimensioning & tolerancing principals considered for design.

5.1.3 **LOCATIONS:-** The Silicone Rubber Composite Insulators are to be used for the 132 KV, 220KV & 400KV lines located near thermal power stations using coal, near sea coast and in the vicinity of cement factories which are getting polluted by certain deposits from the sea, salt deposits in the water vapour from cooling towers coal dust & cement/Fly ash. The clause of pollution is very heavily polluted and creepage distance required is 31 mm /KV.

6.0 DESIGN AND MATERIAL REQUIRMENT

6.1 Core:

The core shall be glass-fibre reinforced epoxy resin rod (FRP) of high strength. Both, glass fibre and resin shall be optimized in the FRP rod. Glass fibres with low content in alkalies shall be boron free E glass or Boron free electrically corrosion resistance (ECR) glass. Use of resin with hydrolysis trend due to water penetration should be prevented i.e. matrix of the FRP rod shall be Hydrolysis resistant. Suitability of Epoxy matrix as well as interface between matrix and fibres is to be considered as design parameter to prevent brittle fracture. The FRP rod should be void free and shall be manufactured through Pultrusion process.

6.2 Housing:

The core of the composite insulator shall be completely covered by a continuous housing consisting of a sheath-weather shed. For moulding of entire weather shed structure on the rod in a one shot moulding process to be employed to avoid multiple i.e. metal fittings may be installed on the rod prior to moulding of the shed controlling moulding lines.

The base polymer shall be 100% Silicon Rubber prior to the addition of reinforcing fillers.

The thickness of compounding material on core should be minimum 3 mm. Manufacturer should furnish a description of its Quality Assurance Programme including fabrication, testing and inspection for any material (i.e. rubber), components (i.e. rod) or hardware (i.e. end fittings) The manufacturer has had fabricated by others should also be included.

Manufacturing methods and material composition documentation will be a part of Technical Bid to be submitted along with offer.

Insulator should have hermatically sealed structure in which the housing material is moulded to cover the interface between the end fitting and the FRP rod. This seal should never be broken during testing or otherwise.

6.3 End fittings:

The composite insulators shall be socket and ball type with the necessary coupling arrangement such that pin shall move freely in the socket but do not get disengaged while in service under various operating and atmospheric conditions.

The socket & ball type metal end fittings shall be designed to transmit the mechanical load to the core & the end fittings shall maintain uniform and consistent mechanical strength Material and methods used in the fabrication of metal parts shall be selected to provide good toughness and ductility. Metal end fittings shall be made from a quality malleable case iron or forged steel or Spheroidal Graphite Iron (SGI) and shall be hot dipped galvanized in accordance with IS 2629. Metal end fittings shall be uniform and without sharp edges or corners and shall be free of cracks, flakes, slivers, slag, blow-holes shrinkage defects and localized porosity. The attachment to the FRP rod shall be performed with a symmetrically controlled crimping method control by acquistic method that compresses the metal radially onto the rod without damage to the rod fibres or resin matrix while providing a strength equal to or greater than the defined and specified ultimate strength to the insulator.

The material used in fittings shall be corrosion resistant.

Nominal dimensions of the pin, ball and socket interior shall be in accordance with the standard shown at CI. No. 4 No joints in ball & socket or pin will be allowed. Outer portion of ball or socket should be Zinc sleeved with minimum 99.95% purity of electrolytic high grade Zinc.The finished surface shall be smooth and shall have a good performance

The surface shall not crack or get chipped due to ageing effect under normal and abnormal service conditions or while handling during transit or erection.

The design of the fittings and the insulators shall be such that there is no local corona formation or discharges likely to cause the interference to either sound or vision transmission.

6.4 GRADING RINGS:

Grading rings shall be provided when system voltages are equal to or greater than 220 KV. For 220KV transmission, grading ring is to be provided at energized end only. For 400 KV transmission, grading ring is to be provided at both ends of an insulator.

All grading rings and brackets shall be designed as an integral part of the insulator assembly with a positive mounting system that allows mounting in only one position. The design of the grading ring shall be such that ring can only be mounted with its orientation towards the weathersheds for maximum RIV and corona control. Grading rings shall be designed in such a manner that the rings can be readily installed and removed with hot line tools without disassembling any other part of the insulator assembly.

Grading ring height (is the distance from the end of the end fitting to the top of corona ring) should be so selected that maximum field minimizes & uniformly distributed along the insulator. Manufacturer should provide reports of successful electrical field modeling testing for the specific insulator design. The EFM should be three dimensional with results containing drawing depicting the electric field in various colours, each of a different voltage level. The result of this study should show that the voltage field surrounding the composite insulator is optimum along the entire length of insulator, with the effected hot end of the insulator being a critical location. The threshold at which corona may or may not be present should be defined as a figure in KV/mm for the designed insulator.

7.0 VERIFICATION OF HOUSING MATERIAL

The manufacturer should provide written verification about housing material, for which base polymer shall be 100% Silicon Rubber prior to the addition of reinforcing fillers considered will provide satisfactory performance in the particular environment mentioned at CI. No.3 It shall meet following requirements

Be homogenous, impermeable, with no fissures, bubbles and strange materials inclusions.

Be designed in order to avoid to avoid formation of localized discharges and to prevent interfaces humid penetration.

Be resistant to corona, KV radiation, ozone, atmospheric contamination, water penetration and power ascs.

8.0 BALL AND SOCKET DESIGNATION

The dimensions of the Ball and Socket shall be 16mm designation upto 90KN and 20mm designation for 120KN & 160KN insulators in accordance with the standard dimensions stated in IEC:120/IS:2486(Part-II)

9.0 DIMENSIONAL TOLERANCE OF COMPOSITE INSULATORS:

The tolerance on all dimensions e.g. diameter, length and creepage distance shall be allowed as follows:

(0.04 d +1.5) mm when d $\leq$ 300 mm.
(0. 025. d +6) mm when d > 300mm.

Where d being the dimensions in millimeters for diameter, length or creepage distance as the case may be.

However, no negative tolerance shall be applicable to creepage distance.

9.1 The Bidder have to submit offer for Silicone Rubber Composite Insulators as per the following parameters.

System Voltage	Mechanical failing load	Creepage distance (mm)		Sectional length (mm)
		Total	Minimum	
132 KV	70 KN	4495	2300	1305
	120 KN	4495	2300	1450
220 KV	70 KN	7595	3800	1885
	120 KN	7595	3800	2030
400 KV	120 KN	13020	6505	3335
	160 KN	13020	6505	3910

10.0 INTGERCHANGEABILITY:

The composite insulators including the ball socket connections shall be standard design suitable for use with the hardware fittings of any make conforming to relevant IEC standards.

11.0 CORONA AND RI PERFORMANCE:

All surfaces shall be clean, smooth, without cuts, abrasions or projections. No part shall be subjected to excessive localized pressure. The insulator and metal part shall be so designed and manufactured that it shall avoid local corona formation and not generate any radio interference beyond specified limit under the operating conditions.

12.0 MARKINGS:

Each insulator shall be legibly marked with the following details as per IEC – 61109.

a. Name or trademark of the manufacturer.
b. Voltage and Type.

c. Month and year of manufacturing.
d. Minimum failing load/ guaranteed mechanical strength in kilo Newton followed by the word 'KN' to facilitate easy identification.
e. Country of manufacture.

13.0 PACKING:

All insulators shall be packed in strong corrugated box of minimum 7 ply duly paletted or wooden crates. The gross weight of the crates along with the material shall not normally exceed 100 kg to avoid handling problem. The crates shall be suitable for outdoor storage under wet climate during rainy season.

The packing shall be of sufficient strength to withstand rough handling during transit, storage at site and subsequent handling in the field.

Suitable cushioning, protective padding, or Dunn age or spacers shall be provided to prevent damage or deformation during transit and handling.

All packing cases shall be marked legibly and correctly so as to ensure safe arrival at their destination and to avoid the possibility of goods being lost or wrongly dispatched on account of faulty packing and faulty or illegible markings. Each wooden case / crate corrugated box shall have all the markings stenciled on it in indelible ink.

The bidder shall provide instructions regarding handling and storage precautions to be taken at site.

14.0 INSPECTION, TESTS AND STANDARDS:

14.1 Proto type or Design or Type: To evaluate core material, housing material, core assembly (core & end fittings), interfaces and connections of sample insulators.

Inspection includes the performance of acceptance, type and design tests. APTRANSCO reserves the right to carry out design and type tests to check conformity of the material with the proto type unit previously approved.

APTRANSCO reserves the right to attend the tests and perform inspections in any stage of the supply, appointing its inspectors and following the approved manufacturing schedule. Inspection and tests scheduled to happen during manufacture shall have their dates informed to APTRANSCO at least 10 days in advance.

The manufacture shall assure APTRANSCO's inspector the right to being fully acquainted with installations and apparatus, check calibrations, is present at the test, check results and in case of doubt, perform new inspections and claim the repetition of any test.

14.2 No material shall be dispatched from its point of manufacture before it has been satisfactorily inspected, tested, and necessary dispatch instructions are issued in writing, except for the cases where waiver of inspection is granted by competent authority of the Purchaser, and even in this case also written dispatch instructions will be issued. Any dispatches before the issue of Dispatch Instructions in writing will be liable for rejection and non-acceptance of the materials by the consignee.

14.3 The acceptance of any quantity of material shall in no way relieve the Bidder of any of his responsibilities for meeting all requirements of the specification, and shall not prevent subsequent rejection if such material is later found to be defective.

14.4 The sample taken from any numbers of crates for carrying out any type of tests will be to the suppliers account.

14.5 TYPE TEST

14.5.1 The type, acceptance, routine tests, any tests specifically demanded by the Purchaser and tests during manufacture shall be carried out on the Insulators free of cost. The test reports shall be in accordance with the socket cap material offered. The original copies of relevant IEC / IS codes with latest revisions/amendments for reference make it available to AP TRANSCO reperentives during acceptance inspection tests and shall \be provided as soft copies

14.5.1.2 Type tests shall mean those tests, which are to be carried out to prove the process of manufacture and general conformity of the material to this specification. These tests shall have to be carried out at the Government Approved Testing Laboratory. Purchaser reserves the right to specify the name of the laboratory also, if so felt. The Type test reports shall not be older than Five years(as per CEA regulation) and shall be valid till validity of offer.
The sample selection will be done as per Clause 14.9 and all the type test as per Clause 14.5.3 and 14.5.4 of this specification, are to be repeated on the offered lot without any extra cost to APTransco on random sample selected by AP TRANSCO engineer at CPRI/ERDA or any NABL approved lab (as decided by APTRANSCO),in the presence of APTRANSCO engineer.

14.5.1.3.1 Acceptance Tests shall mean those tests, which are to be carried out on samples taken from each lot offered for pre-dispatch inspection, for the purposes of acceptance of that lot. These tests shall be carried out at the manufacturers works in presence of Purchaser's representative before the dispatch of the materials to the site.

14.5.1.4 Routine Tests shall mean those tests which are to be carried out on each of the Insulator to check requirements which are likely to vary during production. These tests shall be carried out by the manufacturer on each Insulator and shall have to furnish these reports to the Purchaser's representative during his visit for acceptance tests.

14.5.1.5 Tests during manufacture shall mean those tests, which are to be carried out during the process of manufacture and end inspection by the supplier to ensure the desired quality of the end product to be supplied by him, including all Quality Control checks and Raw Materials testing.

14.5.1.6 The standards to which these tests will be carried out are listed against them. Where a particular test is a specific requirement of this specification, the norms and procedures of the test shall be as specified as mutually agreed between the Bidder and the purchaser in the Quality Assurance Programme.

14.5.1.7 For all type and acceptance tests, the acceptance value shall be the values guaranteed by the Bidder in the "Guaranteed Technical Particulars", of his proposal or the acceptance value specified in this specification, whichever is more stringent for that particular test.

14.5.2 **On the complete composite Insulator with Hardware Fittings:**

(a) Power frequency voltage withstand test with corona control rings/grading ring and arcing horns under wet condition-IEC:383-1993
(b) Impulse voltage withstand test under dry condition.-IEC:383-1993
(c) Wet switching Impulse withstand voltage. IEC :61109-1992
(d) Salt-fog pollution withstand test-Annexure-A
(e) Grading device test-Applicable to 220KV and above voltage class
(f) Electrical Field Modeling test (EFM)-Applicable to 220KV and above voltage class
(g) Power arc test
(h) Wet power frequency test

All the above type test shall be conducted on Single 'I' suspension and Double tension insulator along with hardware fittings.

14.5.3 On Composite Insulator Units.

(a) Tests on interfaces and connections IEC:61109-1992
  i) Dry Power Frequency Voltages Test
  ii) Pre stressing : Sudden Load Release Test
  iii) Thermal Mechanical Test
  iv) Water immersion
  v) Steep Front Impulse Voltage Test
  vi) Verification Test
  vii) Visual examination

(b) Assembled Core Load – Type Tests-IEC:61109-1992
  (i) Average Failing Load of the Core of the assembled Insulator
  (ii) Control of the slope of the strength-time curve of the Insulator

(c) Test of Housing IEC: 61109-1992
(i) Tracking and Erosion test.
(ii) Hardness test
(iii) Accelerated weathering test
(iv) Flammability test

(d) Test for the Core Material IEC:61109-1992
(i) Dye Penetration Test
(ii) Water Diffusion Test

(e) Brittle fracture resistance test – Annexure-A

(f) Multi stress test for 5000 hours as per Annexure- C-IEC:1109

(g) Mechanical load time test IEC:61109-1992 Clause 6.4

14.5.4 On Silicone material

(a) Flammability test IEC: 61109-Amd. 1 or Test as per UL94.
(b) Recovery of Hydrophobicity test-Annexure-A

14.6 Sample Tests (Acceptance Tests) -

When specified on a purchase order, sample tests shall be performed per ANSI C29.11& IEC:61109-1992.

(a) Verification of Dimensions
(b) Verification of Locking System-applicable only in the event ball and socket insulators is specified.(end fittings)
(c) Mechanical Load test- In process testing used to verify the mechanical system is acceptable.(Verification of tightness of the interface between end fittings & insulator housing and of the specified mechanical load, SML)
(d) Galvanizing Test
(e) Silicon Content test (One sample per lot of 200 nos & two samples for lot above 200 nos of 70KN/120KN/160KN SRC insulators shall be taken and test has to be performed through FT-IR (Fourier Transform Infrared) Spectroscopy & TGA (Thermo gravimetric Analysis) or any other approved/acceptable method. Silicon content should not be less than 35.1 %.)
(f) Hydrophobisity Test (In- house test) as per IEC -61109/2008 (The surface of the selected sample shall be cleaned with Isopropyl alcohol. Allow the surface to dry and spray with water. Record the HC classification)
(g) Dye Penetration test (In house test) on FRP rod as per IEC – 61109/2008
(h) Ensuring Permanent Embossing: Permanent embossing of Serial No, Batch No, Month/year and APTRANSCO

14.7 Routine Tests:

The Following tests shall be performed on every insulator produced as per IEC:61109-1992.

(a) Mechanical Test: Every insulator shall withstand for a period not less than 10 seconds a tensile load equal to or greater than its Routine Test Load (50%of the Specified Mechanical Load)
(b) Visual Examination: Every insulator shall be examined to ensure its conformance to the manufacturer's drawing. Superficial polymer surface defects of an area less than 25 square millimeters (total area not to exceed 2% of total insulator surface area) and depth less than 1 mm shall be acceptance.

14.8 Additional Tests

14.8.1 The Purchaser reserves the right of getting done any other test(s) of reasonable nature carried out at Purchaser's premises, at site, or in any other place in addition to the aforesaid type, acceptance and routine tests to satisfy himself that the material comply with the specifications. In such case all the expenses will be to Suppliers account.

14.9 Sample Batch For Type Testing

14.9.1 The Bidder shall offer at least 10% of the ordered quantity or 300 nos. whichever is higher, for selection of samples required for conducting the entire type test.

14.9.2 The Bidder is required to carry out all the acceptance tests successfully in the presence of Purchaser's representative before dispatch of the selected sample to the testing laboratory for type test.

15.0 TEST REPORTS

15.1. Copies of type test reports shall be furnished in at least two (2) copies along with one original. One copy shall be returned duly certified by the Purchaser only after which the material already inspected i.e. the Materials manufactured for selection of sample for type test, shall be dispatched on receipt of Dispatch Instructions from the Chief Engineer/Lift Irrigation, APTRANSCO, Vidyut Soudha, Vijayawada.

15.2 Record of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the purchaser's representative.

15.3 Test Certificates of test during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when desired by the Purchaser.

16.0 TEST FACILITIES

16.1 The following additional facilities shall be available at Supplier's works:-

a) Calibration Reports from Government approved testing laboratory of various testing and measuring equipment including tensile testing machine, resistance measurement facilities, burette, thermometer, barometer etc.

b) Finished insulator shall be checked for dimension verification and surface finish separately.

c) The bidder should have all the routine and acceptance testing facilities, in house**.**

16.0 QUALITY ASSURANCE PLAN

16.1 The bidder shall invariably furnish following information along with his offer, failing which his offer shall be rejected.

i) Statement giving list of important raw materials, proposed to be used in the manufacture of the insulator against this Specification, names of sub suppliers for the raw materials, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials in presence of Bidder's

representative as routine and/or acceptance during production and on finished goods, copies of test certificates.

ii) Information and copies of test certificates as in (i) above is respect of bought out accessories.

iii) List of manufacturing facilities available.

iv) Level of automation achieved and lists of areas where manual processing exists.

v) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

vi) List of testing equipment available with the Bidder for final testing of Insulator specified. In case if the Bidder does not possess all the Routine and Acceptance testing facilities the tender will be rejected.

vii) The Purchaser reserves the right for factory inspection to verify the facts quoted in the offer. If any of the facts are found to be misleading or incorrect the offer of that Bidder will be our rightly rejected and he may be black listed.

viii) Special features provided to make it maintenance free.

ix) Bidder shall also submit the Field Quality Plan (FQP) along with Technical Bid.

16.2 The bidder shall also submit following information to the purchaser along with the technical Bid.

i) List of raw materials as well as bought out accessories, and the name of suppliers of raw materials as well as bought out accessories.

ii) Type test certificates of the raw material and bought out accessories.

iii) Quality assurance plan (QAP) with hold points for purchaser's inspection.

16.3 The Bidders shall submit the routine test certificates of all the bought out items, accessories etc.

17.0 **GUARANTEED TECHNICAL PARTICULARS:**

Full guaranteed particulars including dry and wet flashovers, puncture and impulse voltages, corona formation voltages, creepage distances, length, voltage distribution on strings etc., shall be furnished.

ANNEXURE - I

REQUIRED RATING AND ELECTRICAL CHARACTERISTICS

Sl. No.	Characteristics	For Silicone Rubber Composite Insulators string					
		132 KV		220 KV		400 KV	
		70 KN	120 KN	70 KN	120 KN	160 KN	120 KN
1.	Type description	Silicone Rubber Composite Insulators suitable for B95 couplings at the line & tower ends.				B&S	
2.	Colour & surface of rubber portion.	Grey					
3.	Ferrous parts	IEC 61109 & ASTM A 153					
4.	Check clip provided In the socket.	'R' Clip					
5.	Ball pin	Drop forged steel					
6.	Socket fitting	Malleable Ductile Iron					
7.	Ball pin designation	16m mB	20 mm	16 mm B	20mm	20mm	20 mm
8.	a) Diameter and spacing of	AS PER IEC 61109					
	b) Tolerances						
9.	Minimum failing Load of insulator (strings)	7000 kg	11500 KG	7000 kg	11500 KG	160 KN	120 KN
10.	Minimum total creepage distance in mm	4495	4495	7595	7595	13020	13020
11.	Minimum protected creepage distance in mm	2300	2300	3800	3800	6505	6505
12.	Dry Arcing Distance mm min.	1100	1100	1900	1900	3650	3650
13.	Combined Mechanical and Electrical strength in kg	7000 kg	11500 KG	7000 kg	11500 KG	160 KN	120 KN
14.	One minute power Frequency withstand voltage (kV) Dry	275	275	460	460	630 KV	630 KV

14 (a)	Sectional length	1305	1450	1885	2030	3910	3335
15.	One minute power Frequency withstand voltage (kV) Wet	275	275	460	460	630 KV	630 KV
16.	One minute power frequency Flashover voltage Dry/Wet (KV)	275	275	> 460	> 460	>630	>630
17. i)	AC wet withstand voltage : KV	275	275	460	460	630	630
ii)	L.I. Pos with stand voltage : KV	650	705	1050	1050	1550	1550
iii)	S.I. Wet WSV : KV	Not applicable				1050	1050
18.	Visible discharge voltage (kV)	90	90	140	140	320 KV	320 KV
19.	Impulse flash over voltage (kV) (min):						
	Positive	650	650	1050	1050	1550	1550
	Negative	710	710	1100	1100	1600	1600
20.	Flashover Power Frequency voltage (kV)						
	Dry	270	270	460	460	630	630
	Wet	270	270	460	460	630	630
21.	Standards according to which the rubber insulator shall be manufactured and tested.	IEC 61109/ ANSI C 29-11, 29 – 12					
22.	Mechanical breakings strength.	7000 kg	11500 KG	7000 kg	11500 KG	160 KN	120 KN
23.	Routine Mechanical Load	60 KN	80 KN	60 KN	80 KN	120 KN	90 KN
24.	Net weight of strings (approx).	5.5 KG	6 KG	7 KG	9 KG	21 KG	24 Kg

Annexure - II

GUARANTEED PERFORMANCE & OTHER PARTICULARS

(While furnishing these details, the values in Annexure-I also are to be furnished.)

Sl. No.	Description	For Silicone Rubber Composite Insulators string					
		132 KV		220 KV		400 KV	
		70 KN	120 KN	70 KN	120 KN	120 KN	160 KN
(1)	(2)	(3)	(4)	(5)	(6)	(7)	

1. a) Type of insulator (Ball and socket)

   b) Size and designation ball and socket with standard to which it will conform.

2. Maker's name, address and country.

3. Dimensions:

   Diameter in mm
   Unit spacing in mm.
   Creepage distance in mm.

   i) Total
   ii) Protected

4. Mechanical Values:

   Combined Mechanical and Electrical strength in KGs.

   Mechanical breaking strength in KGs.

   Routine Mechanical test load (KGs.)

5. Electrical Values:

   Power frequency Flashover Voltage DRY (KV)

   Power frequency Flashover Voltage WET (KV)

   Impulse flashover voltage 1.2 x 40 mircro second negative wave (KV)

	Impulse flashover voltage 1.2x40 mircro Second negative wave (KV)	
	Power frequency withstand voltage : DRY (KV)	
	Power frequency withstand voltage : WET (KV)	
	Impulse withstand test voltage 1.2x40 micro second positive and negative wave (KV)	
	Power frequency puncture Voltage (KV)	
6.	Net weight of insulator unit in KGs.	
7.	Strandard specification to which insulators will conform.	
8.	Other particulars if any.	

==========

SIGNATURE OF THE TENDERER